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Research Paper

Development of a Lightweight Interpretable Model for Early Stage Detection of Child Grooming Indicators in Short Conversational Text

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ABSTRACT

Early detection of online child grooming requires systems that recognise contextual risk in short messages while limiting false alarms, protecting sensitive communications, and operating within available hardware. Rare predatory examples, gradual escalation across conversations, and threats conveyed through images make this task difficult. This study develops Anecdote, a local multimodal safeguarding pipeline combining Qwen3 8B through Ollama, NudeNet for explicit image detection, and EasyOCR for embedded text extraction. The text corpus contains 68,007 extracted messages, including 2,015 predatory examples. The methodology uses backtranslation and a weighted binary cross entropy objective to address class imbalance. The implementation adds input sanitisation, prompt injection indicators, lazy loading of vision models, session risk tracking, and structured outputs for moderation. In a curated evaluation of 30 text cases, thresholds of 0.50 and 0.60 achieved precision of 1.0000, recall of 0.8667, and F1 of 0.9286; reported text ROC AUC was 0.9956. For 16 image cases, a threshold of 0.60 achieved precision of 1.0000, recall of 0.5455, and F1 of 0.7059, with ROC AUC of 0.7727. Mean latency was 2.142 seconds for text and 2.407 seconds for images, exceeding the respective 50 and 100 millisecond targets. These results support local integration and preliminary threshold calibration, while larger independent evaluations and faster inference remain necessary.

Keywords: online child grooming; short text classification; multimodal safeguarding; local inference; threshold calibration.

INTRODUCTION

Motivation

Online child grooming involves building trust with a child to enable sexual exploitation. Messaging services, social media, and gaming platforms provide channels for repeated private contact, secrecy requests, coercion, and attempts to isolate children from trusted adults. These behaviours can develop gradually and may initially resemble ordinary conversation (Chiu & Quayle, 2022; Lorenzo-Dus, 2023). Detection therefore depends on the relationship between words and context, rather than the presence of isolated keywords.

User reports and retrospective investigations are important safeguards, but they may begin after harmful exchanges have progressed. Situational crime prevention provides a rationale for intervention during an unfolding interaction: making offending harder and increasing the chance of detection within the environment in which it occurs (Cornish & Clarke, 1986; Seto, 2021). A message screening service could support warnings and moderation review before a conversation escalates. The present study evaluates the detection component; it does not measure whether alerts prevent abuse.

Local inference is also relevant because the communications being assessed are sensitive. Running the models within the hosting environment can reduce reliance on external moderation services. However, contextual language processing must then fit available memory and provide a response quickly enough to support the intended workflow. Anecdote examines this combination of detection, integration, and deployment constraints.

Problem Definition

The computational task is binary risk classification of short conversational messages and static image payloads. For an incoming item, the system produces a risk score between zero and one, associated categories, an explanatory rationale, and a moderation flag. Image analysis combines explicit visual content detection with text recovered through optical character recognition. Session state allows repeated risk indicators to influence a later decision.

Four requirements shape the problem. The classifier must identify rare predatory examples in a heavily imbalanced corpus; distinguish concerning intent from ordinary references to age, school, affection, or photographs; incorporate image and session signals; and operate locally with explicit handling of inference failures. The score is a model assessment of indicators, rather than a verified probability that an individual is an offender.

The scope covers text and static images. Audio, video, mobile application development, and formal reporting to law enforcement fall outside the study. Multilingual capability is a design objective, but the experiments presented here do not establish performance separately for individual languages.

Research Contributions

The study contributes a modular local pipeline that brings language reasoning, explicit image screening, and OCR into one moderation interface. Its interchangeable text backend and vision models loaded on demand support experimentation without tightly coupling the orchestration logic to a single model implementation.

It also combines single message assessment with session risk history, input sanitisation, prompt injection indicators, and degraded outputs when components fail. Structured risk categories, rationale, and flag reasons provide information for a reviewer to inspect, although they do not constitute a validated explanation of the model's internal decision process.

Finally, the study reports threshold sweeps for text and image classification together with latency, throughput, RAM, and GPU memory measurements. These experiments show the sensitivity gained by lowering thresholds and identify language model inference as the principal performance bottleneck.

BACKGROUND AND RELATED WORK

Existing Approaches

Traditional text classifiers use representations such as term frequency inverse document frequency with algorithms including support vector machines, random forests, and Naive Bayes. These methods provide practical baselines, but their representations may overlook implicit manipulation and the contextual meaning of informal language (Borj et al., 2023; Minaee et al., 2021).

Transformer models use attention to represent relationships between tokens, providing a basis for contextual text classification (Vaswani et al., 2017; Devlin et al., 2019). The related work examines compact encoders and instruction tuned language models, including the tradeoff between contextual capacity and inference cost (Fleron, 2021; Mylonas et al., 2025). Contrastive training is considered as a way to separate benign informal language from concerning patterns (Schittenhelm et al., 2024). These approaches motivate contextual modelling, but their reported results are not interchangeable with the small case evaluation used in this study.

Linguistic and temporal approaches model grooming as a process involving trust building, isolation, boundary testing, and solicitation (Lorenzo-Dus, 2023; Zhang et al., 2025). Such approaches can preserve progression across a conversation, while single message classifiers can produce an earlier assessment. Anecdote combines immediate item scoring with a limited session risk history, rather than treating either view as sufficient on its own.

Other work addresses minority class augmentation, quantisation, and decision threshold calibration (Waezi & Fani, 2025; Al-Mansoor & Rodriguez, 2026; O'Connor et al., 2026). Backtranslation varies the phrasing of existing examples; weighted loss changes the penalty for missing the minority class; and threshold adjustment changes the operating point after scoring. Each acts at a different stage and should be evaluated separately to establish its contribution.

Table 1 Approaches discussed in the related work

Approach	Primary focus	Constraint relevant to this study
Feature based classifiers	Word and statistical features	Limited representation of implicit context
Transformer encoders	Contextual sequence classification	Requires task specific training and validation
Instruction tuned language models	Context and structured rationales	Generation latency and memory demand
Temporal linguistic models	Escalation over multiple turns	Dependence on conversation history
Augmentation and weighted loss	Rare class learning	Semantic drift and evaluation leakage risks
Quantisation and threshold calibration	Deployment and operating point	Accuracy and alert volume tradeoffs

Research Gaps

Three gaps guide the design. First, a text only detector cannot directly assess explicit visual content or language embedded in an image. A common orchestration layer is needed to route each modality and return a consistent decision. Second, early message assessment and conversation level progression require different information. A detector that ignores history may miss cumulative risk, while one that requires extensive history may delay an initial warning.

Third, local deployment introduces constraints that classification scores alone do not describe. Model loading, quantisation, memory consumption, inference time, and component failures affect whether a detector can support a live moderation workflow. Anecdote addresses these issues through an integrated prototype and measures the resulting tradeoffs. Its experiments do not establish superiority over other trained model architectures.

METHODOLOGY

Proposed Method

Anecdote is a Python package organised around an orchestrator, a text backend interface, a vision subsystem, and a risk engine. The command line entry point accepts a message or image and directs it to the relevant processing path. The core modules handle classification and session state, while setup utilities manage environment and model availability. This separation allows a model backend to be replaced without rewriting the decision workflow.

For text input, the pipeline removes hidden Unicode characters and checks for patterns associated with prompt injection. Suspicious input is marked in the output and receives a minimum risk score of 0.75. The sanitised message is assessed by the local Qwen3 8B backend, which returns a score, categories, and a short rationale. The runtime interface supports the Ollama implementation, a deterministic testing stub, and a placeholder for additional backends.

For an image, the loader validates the input before invoking NudeNet and EasyOCR. NudeNet supplies explicit content scores; EasyOCR extracts any embedded text, which is passed through the text assessment path. Both vision tools load their model weights on first use so that text only sessions avoid unnecessary vision initialisation. Figure 1 summarises the processing paths and their shared decision stage.

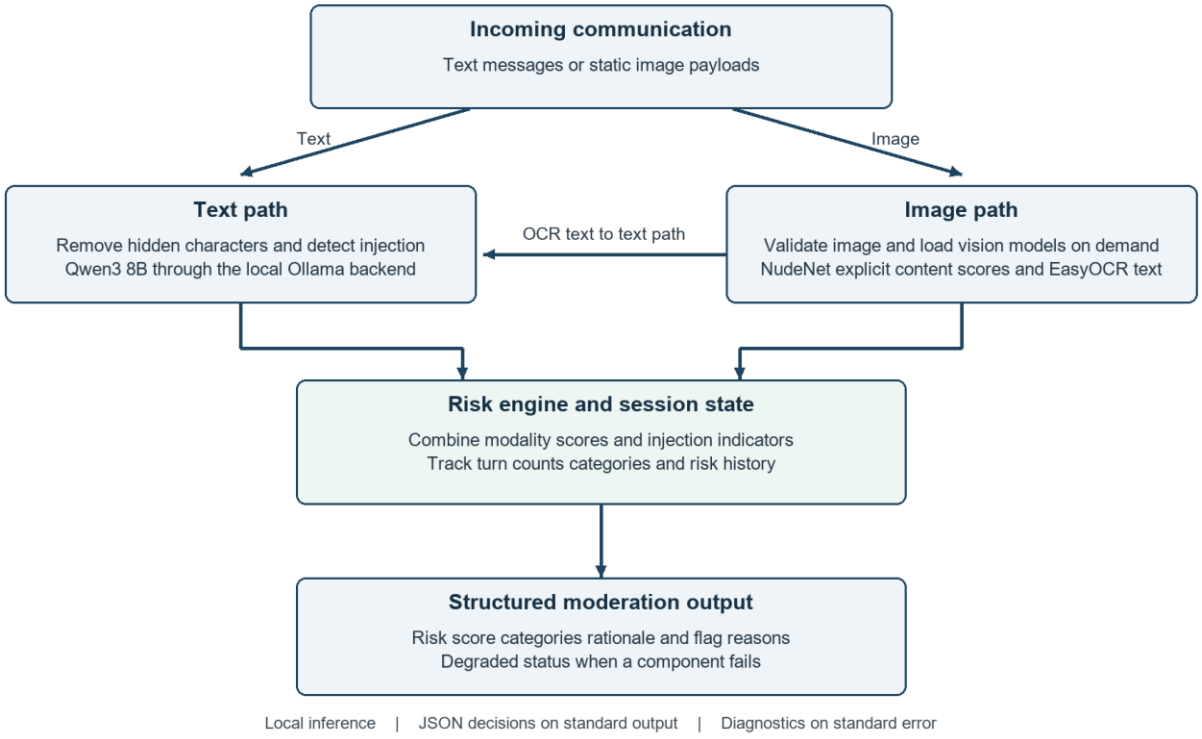


Figure 1 Anecdote local multimodal architecture and shared risk assessment

The risk engine considers modality scores, injection indicators, and session state. The configured session rules flag a current risk score at or above 0.85, or repeated historical scores at or above 0.70; the configuration specifies two occurrences for the history rule. These session rules are distinct from the classification thresholds swept in the experiments. The threshold study therefore describes score based classification performance and should not be read as an evaluation of every session rule.

Session state is keyed by session identifier and tracks turn counts, risk history, category flags, and a rolling context summary. Local in memory processing reduces external transmission, but privacy

depends on the content retained in summaries and outputs as well as on where inference occurs. The implementation provides structured decisions on standard output and diagnostics on standard error to support integration with other software.

If model inference times out, parsing fails, or an image cannot be processed, the orchestrator returns a degraded result with an elevated score and a manual review category. In the example text backend fallback, the risk score is 0.60. A consuming application must inspect the degraded status and review category directly, because a fallback score alone need not cross the normal 0.85 session flag threshold.

Data and Models

The text data comprise 68,007 single message records extracted from PAN12 XML chat material and predatory chat logs. Of these, 65,992 are benign and 2,015 are predatory, giving a predatory prevalence of approximately 2.96 percent. Extraction turns complete chat records into the short message units used by the service. Cleaning removes tags, timestamps, and system metadata while retaining emojis, slang, punctuation, and non English words.

Table 2 Text corpus composition and stated partitions

Data component	Messages	Share or purpose
Benign records	65,992	97.04 percent
Predatory records	2,015	2.96 percent
Total extracted corpus	68,007	100 percent
Training partition	47,605	Approximately 70 percent
Validation partition	10,201	Approximately 15 percent
Test partition	10,201	Approximately 15 percent

The methodology describes translating predatory English messages into German, French, and Spanish and back into English. This is intended to increase phrasing variation in the minority class. It then specifies a stratified 70/15/15 split and a Qwen tokenizer limit of 128 tokens per message. The stated partition counts sum to the original 68,007 records; the size of an expanded augmented corpus is not separately established. Consequently, the partition totals and augmentation procedure should be treated as distinct reported details.

Qwen3 8B supplies the text risk assessment through Ollama. The described training objective is weighted binary cross entropy, with additional weight on positive examples to penalise missed predatory messages. The training workflow specifies PyTorch and Hugging Face Transformers, AdamW, a linear learning rate schedule, and checkpoint selection. The objective is:

$$L = -(1/N) \Sigma_i [w_1 y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Here N is the number of examples, y_i is the binary target, p_i is the predicted positive score, and w_1 is the positive class weight. Reproducing the training stage requires the numerical weight, learning rate, batch size, epoch count, random seed, checkpoint identifier, and details linking the loss to the deployed scoring model.

For deployment, the methodology describes 4 bit and 8 bit GGUF quantisation for Qwen3 8B, INT8 ONNX execution for NudeNet, and PyTorch execution for EasyOCR. EasyOCR combines text region detection with character recognition to recover image text. The experiments do not isolate the effect of quantisation or report a separate OCR accuracy benchmark.

Experimental Setup

The reported environment is a cloud Linux server running Python 3.13 with an NVIDIA Tesla T4 and nominally 16 GB GPU memory. The software specification lists PyTorch 2.6, Hugging Face Transformers 4.41.0, ONNX Runtime 1.20, EasyOCR 1.7, NudeNet 2.0, and Ollama. The language model is accessed through the local endpoint at localhost port 11434 with GPU acceleration enabled. Local here refers to inference within the hosting environment, which in these experiments is a cloud server.

The classification experiments use 30 curated text cases, evenly divided between benign and predatory messages, and 16 image cases containing 5 benign and 11 explicit or predatory examples. These case sets support the reported threshold sweeps. They are separate from the stated 10,201 message test partition, for which a complete set of classification results is not presented.

The thresholds evaluated are 0.50, 0.60, 0.70, 0.75, 0.80, 0.85, and 0.90. Text at 0.70 provides the reported baseline configuration; image thresholds of 0.85 and 0.90 represent conservative defaults. Latency targets are 50 milliseconds for text and 100 milliseconds for images. Execution logging records per item latency, pipeline stages, throughput, process RAM, and GPU memory consumption.

EXPERIMENTS AND RESULTS

Experiments

The evaluation has three components. The text experiment assesses messages involving benign exchanges, isolation, solicitation, and evasion attempts. The image experiment assesses explicit visual content and images with embedded text. The system experiment measures latency and resource use while these items pass through the local inference pipeline. Threshold comparisons use the same evaluated scores within each modality.

Demonstrations also show structured detection outputs containing the session identifier, component scores, categories, rationale, and flag reasons. Such examples illustrate the interface and decision reporting; the classification counts below provide the quantitative evidence. No independent experiment isolates the contribution of session history, injection handling, backtranslation, or weighted loss.

Evaluation Metrics

A true positive (TP) is a correctly detected positive case; a false positive (FP) is a benign case flagged as positive; a true negative (TN) is a correctly accepted benign case; and a false negative (FN) is a missed positive case. Precision is TP/(TP + FP), recall is TP/(TP + FN), and F1 is 2TP/(2TP + FP + FN). Specificity is TN/(TN + FP).

Balanced accuracy is the mean of recall and specificity: balanced accuracy = 0.5 × [TP/(TP + FN) + TN/(TN + FP)]. Values in the tables are calculated from the confusion counts and rounded to four decimal places. ROC AUC summarises ranking across thresholds and is reported separately for each modality. Precision is undefined when the model makes no positive predictions; F1 is zero when positive ground truth cases exist but none is detected.

Key Findings

For text, thresholds of 0.50 and 0.60 each detect 13 of 15 predatory cases and correctly classify all 15 benign cases. Precision is 1.0000, recall is 0.8667, F1 is 0.9286, and balanced accuracy is 0.9333. Reported ROC AUC is 0.9956. Table 3 shows the full threshold sweep, and Figure 2 shows how raising the threshold reduces recall and F1 while precision remains unchanged in this sample.

Table 3 Text classification results on 30 curated cases

Threshold	TP	FP	TN	FN	Precision	Recall	F1	Balanced accuracy
0.50	13	0	15	2	1.0000	0.8667	0.9286	0.9333
0.60	13	0	15	2	1.0000	0.8667	0.9286	0.9333
0.70	10	0	15	5	1.0000	0.6667	0.8000	0.8333
0.75	10	0	15	5	1.0000	0.6667	0.8000	0.8333
0.80	10	0	15	5	1.0000	0.6667	0.8000	0.8333
0.85	10	0	15	5	1.0000	0.6667	0.8000	0.8333
0.90	4	0	15	11	1.0000	0.2667	0.4211	0.6333

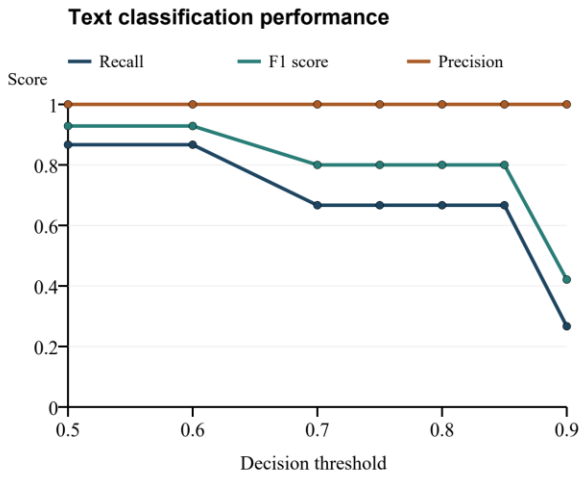


Figure 2 Text precision recall and F1 across thresholds using the counts in Table 3

For images, a threshold of 0.60 detects 6 of 11 positive cases with no false positives among the 5 benign cases. Precision is 1.0000, recall is 0.5455, F1 is 0.7059, and balanced accuracy is 0.7727. Lowering the threshold to 0.50 detects one additional positive case but introduces two false positives. At 0.85 and 0.90, no positive image is detected. Reported image ROC AUC is 0.7727.

Table 4 Image classification results on 16 curated cases

Threshold	TP	FP	TN	FN	Precision	Recall	F1	Balanced accuracy
0.50	7	2	3	4	0.7778	0.6364	0.7000	0.6182
0.60	6	0	5	5	1.0000	0.5455	0.7059	0.7727
0.70	1	0	5	10	1.0000	0.0909	0.1667	0.5455
0.75	1	0	5	10	1.0000	0.0909	0.1667	0.5455
0.80	1	0	5	10	1.0000	0.0909	0.1667	0.5455
0.85	0	0	5	11	Undefined	0.0000	0.0000	0.5000
0.90	0	0	5	11	Undefined	0.0000	0.0000	0.5000

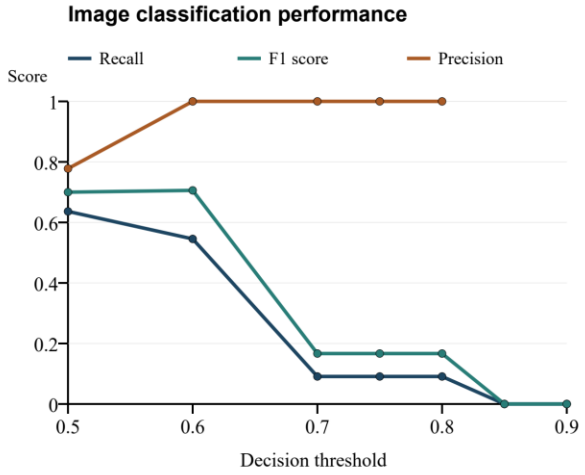


Figure 3 Image performance across thresholds using Table 4 with precision omitted where no positive prediction exists

The runtime results show that model integration is feasible in the measured environment, but the original latency targets are unmet. Mean text processing time is 2,141.917 milliseconds and mean image time is 2,406.971 milliseconds. These are approximately 42.8 and 24.1 times the respective targets. Text throughput is 0.467 messages per second and image throughput is 0.415 items per second.

Table 5 End to end latency and throughput

Measure	Text	Image
Mean latency in milliseconds	2,141.917	2,406.971
Median latency in milliseconds	2,136.104	1,984.402
Minimum latency in milliseconds	1,639.498	424.104
Maximum latency in milliseconds	3,365.520	8,689.757
95th percentile in milliseconds	2,716.187	8,689.757
Throughput in items per second	0.467	0.415
Target latency in milliseconds	50.0	100.0

The mean text sanitisation time is only 0.072 milliseconds, whereas mean language model processing is 2,141.762 milliseconds. The language model therefore accounts for virtually all measured text inference time. These stage measurements identify the main optimisation target; they do not show that every part of image processing has the same bottleneck.

Table 6 Reported memory consumption

Measure	Value
Initial peak process RAM	312.45 MB
Final peak process RAM	1,444.89 MB
Increase in process RAM	1,132.44 MB
Allocated GPU memory	8,771 MB
Available GPU memory	15,360 MB
GPU memory utilisation	57.10 percent

Memory measurements indicate that the evaluated configuration fits within the reported GPU capacity. Process RAM and GPU memory are distinct measurements and should not be added or treated as a complete accounting of every process involved in serving the language model.

Comparisons with Baselines

Against the text baseline at 0.70, lowering the threshold to 0.60 increases true positives from 10 to 13 and reduces false negatives from 5 to 2. Recall improves by 0.2000, or 20 percentage points, while F1 increases from 0.8000 to 0.9286. No additional false positive is observed. The reduction in missed positive cases is 60 percent within this particular test set.

Against the conservative image baseline at 0.85, a threshold of 0.60 raises recall from zero to 0.5455 and balanced accuracy from 0.5000 to 0.7727. Relative to 0.50, the 0.60 threshold trades one detected positive for removal of two false positives. Its F1 advantage is small, rising from 0.7000 to 0.7059, so the choice reflects an operating tradeoff rather than a universal optimum.

These are comparisons between operating thresholds of the same pipeline. The experiments do not include a matched keyword filter, support vector machine, compact transformer encoder, or alternative language model trained and tested under the same protocol. The related work therefore provides context

for future baselines, rather than evidence that Anecdote outperforms those models.

DISCUSSION

Interpretation of Findings

The text results suggest that score ranking is stronger than the conservative operating point: ROC AUC is high, yet a 0.70 threshold misses one third of positive cases. Lowering the threshold recovers three additional positives without adding false alarms in the evaluated sample. This supports careful threshold calibration, but the small set cannot establish that precision will remain perfect on a broader population.

The image results show a different score distribution and a greater sensitivity problem. A high threshold suppresses every positive decision, while even the best observed F1 setting misses 5 of 11 positive images. Explicit visual content detection also addresses a narrower construct than grooming intent. Neither an explicit image flag nor an OCR score alone establishes the nature of the relationship between participants.

A threshold selected using these same cases may perform less well on new data. The evaluated text set is balanced, whereas only 2.96 percent of the extracted corpus is predatory. Precision depends on prevalence and the false positive rate, so zero observed false positives among 15 benign text cases cannot establish a sufficiently low false alarm rate for high volume deployment.

Implications

The architecture demonstrates how local language and vision components can return a common, inspectable moderation result. Categories, rationale, component scores, and degraded status can help a human reviewer understand why an item was escalated. The study supports this operational form of interpretability; it does not validate the faithfulness of generated rationales to the underlying model computation.

Measured latency limits the intended intervention workflow. A service averaging about two seconds per item and fewer than one item per second does not meet the stated inline processing targets. Queue based triage or further optimisation may be practical directions to investigate, but production capacity and concurrent request handling require their own measurements.

Privacy also requires attention to retained content. The implementation schema contains analysed text, and the summary helper retains a fragment of the input. Local inference therefore reduces external transmission but does not justify a claim that the system never retains or returns text. A score only design would need to remove content fragments, define retention and deletion rules, and validate those rules against the complete runtime.

Limitations

The evaluation is preliminary. Thirty text cases and sixteen image cases provide limited coverage of slang, ambiguous intent, demographic variation, and unusual visual inputs. The results do not measure language specific performance, real world alert volume, or robustness across independent platforms. No confidence intervals, repeated runs, or matched model comparisons are provided.

The data preparation procedure places augmentation before splitting. Without grouping original conversations and their translated variants, related content could cross training and evaluation boundaries. A stronger protocol would split by conversation or participant first, augment only the training partition, and preserve validation and test data in their original form. The expanded sample count and complete training configuration also require documentation for reproducibility.

The runtime measurements describe one GPU environment. They do not establish CPU performance, performance on lower memory devices, concurrency limits, or the difference between cold and warm execution. Lazy loading is implemented, but an ablation comparing eager and lazy loading is needed to quantify its savings. Likewise, separate experiments are needed to measure OCR errors, injection resistance, session escalation accuracy, and the value of each modelling component.

Finally, code and configuration should be aligned with the stated safeguarding guarantees. A degraded score of 0.60 does not necessarily trigger a session threshold of 0.85, and the supplied prompt describes single message classification while the architecture also discusses conversational context. Testing these paths explicitly is necessary before claiming reliable failure escalation or validated multi turn understanding.

CONCLUSION

Main Contributions

This study develops Anecdote as a local multimodal pipeline for detecting child grooming indicators in short conversational inputs. It integrates Qwen3 8B, NudeNet, and EasyOCR with model abstraction, vision loading on demand, input sanitisation, session risk tracking, and structured decision reporting. These components provide an implementation foundation for moderation research involving text and static images.

The experiments show that lowering text thresholds to 0.50 or 0.60 improves recall to 0.8667 and F1 to 0.9286 in the 30 case evaluation, while a 0.60 image threshold produces recall of 0.5455 and F1 of 0.7059 across 16 images. The results also establish a substantial latency gap: mean execution exceeds two seconds in both modalities. The principal outcome is an integrated prototype with promising preliminary text discrimination and clear requirements for stronger image detection, evaluation, and runtime efficiency.

Future Research Directions

Future experiments should use independent, conversation grouped test sets with realistic class prevalence and broader linguistic coverage. Training only augmentation, documented checkpoints and hyperparameters, repeated trials, and uncertainty estimates would make results more reproducible. Matched keyword, feature based, compact encoder, and alternative language model baselines would establish whether the additional complexity improves detection under the same conditions.

Latency work should prioritise smaller specialised models, classification heads, and optimised local serving. Distillation, TensorRT LLM or vLLM serving, speculative decoding, and progressive scoring are candidates for investigation; their benefits must be measured together with accuracy and memory use. Cold start and concurrent request experiments are needed before selecting a serving architecture.

Further studies should isolate the contributions of OCR, visual screening, injection handling, and session history through ablation experiments. Longitudinal evaluation should test whether risk trajectories identify escalation earlier without creating excessive alerts. Privacy tests should verify content retention and deletion, while failure tests should establish that degraded results consistently reach human review. A subsequent field evaluation could then examine moderator workload and the effectiveness of interventions.

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